

CAPSTONE PROJECT / ENTERPRISE AI RAG SUPPORT

SCHOOLTRUST AI

Verified Parent Support and Knowledge Improvement System

Evidence for the problem, existing baseline and selected innovation gap.

Project	SchoolTrust AI
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Research method	Official product and documentation review plus case-study observation
Case-study evidence	Umeed school chatbot fee and fallback-answer failure
Version / date	Version 1.0 14 July 2026

MARKET VALIDATION AND INNOVATION BRIEF

One focused project. Approved sources. Human-controlled improvement.

1. Validated Problem

Evidence	Finding	Project implication
Umeed case study	The existing school assistant had a knowledge base but still returned a generic office-contact fallback for questions whose answers existed.	Retrieval success and answer quality must be tested, not assumed.
Annual-fund case	One generic amount was incomplete because new admissions and promoted students use different categories.	Critical answers require category clarification and stronger guardrails.
Changing school information	Fees, timings, notices and admissions information can become stale.	Approved sources require effective and expiry dates.
Repeated parent questions	Unsupported questions are usually answered manually but not converted into structured improvements.	Unresolved questions should become prioritised knowledge gaps.

2. Existing Market Baseline

Source	Existing capability	What remains to demonstrate
Zendesk AI Knowledge Base	Identifies content gaps, detects outdated resources and suggests new topics from support demand.	A student implementation still needs a controlled end-to-end workflow that works on school policy data.
Ada knowledge guidance	Treats unresolved AI conversations as evidence of missing knowledge and emphasises authorised sources and freshness.	The capstone must turn this principle into visible gap, approval and effective-date controls.
Intercom content recommendations	Uses unresolved conversations to recommend knowledge improvements ranked by impact.	The school prototype needs a simpler, believable administrator approval loop.
n8n evaluation documentation	Supports running AI workflow test datasets and evaluation logic.	The capstone should prove critical school answers through regression tests.

3. Gap Selected for the Capstone

Selected gap

Small schools need an affordable support workflow that verifies critical public information, records unresolved questions, requires administrator approval, and retests important answers after every knowledge improvement.

Ordinary RAG chatbot	SchoolTrust AI improvement
Retrieves text and generates an answer.	Filters by approval status and effective date before generation.
Uses one confidence approach for every question.	Applies a stronger threshold to fees, dates, timings and policies.
Returns a generic fallback.	Creates or increments a structured knowledge-gap record.
Updates knowledge manually without proof.	Records administrator approval and runs regression questions.
May silently select one conflicting value.	States valid categories separately, asks clarification or escalates.

Ordinary RAG chatbot	SchoolTrust AI improvement
Focuses only on conversation.	Provides a knowledge-health dashboard and test results.

4. Innovation Claim

Honest positioning

SchoolTrust AI does not claim that RAG, content-gap detection or regression testing are individually new. Its capstone innovation is the school-specific combination of critical-fact assurance, date-valid sources, human-approved knowledge-gap recovery and automatic retesting in one student-feasible workflow.

Innovation dimension	Implementation evidence
Problem specificity	Different school categories can produce different valid fee answers.
Safety design	Source identifier validation plus 0.82 critical-answer threshold.
Human control	Administrator token, approval form, audit log and no automatic policy authority.
Learning loop	Repeated gap grouping, approved update, re-embedding and regression evaluation.
Commercial relevance	Reduces repetitive office questions while exposing missing or stale support content.
Student feasibility	One school, three webhooks, one database, one static interface and 38 n8n nodes.

5. Research Sources

Organisation	Source URL
Zendesk	https://www.zendesk.com/service/help-center/ai-knowledge-base/
Ada	https://www.ada.cx/blog/crafting-a-smarter-knowledge-base-the-key-to-ai-customer-service-software-that-works/
Intercom	https://www.intercom.com/help/en/articles/11394959-use-ai-powered-content-recommendations-to-improve-fin
n8n	https://docs.n8n.io/build/integrate-ai/test-and-improve-ai-workflows/understand-why-to-test

Research limitation: This review establishes a credible product gap for a capstone. It is not a patent search and does not support a claim of worldwide novelty.